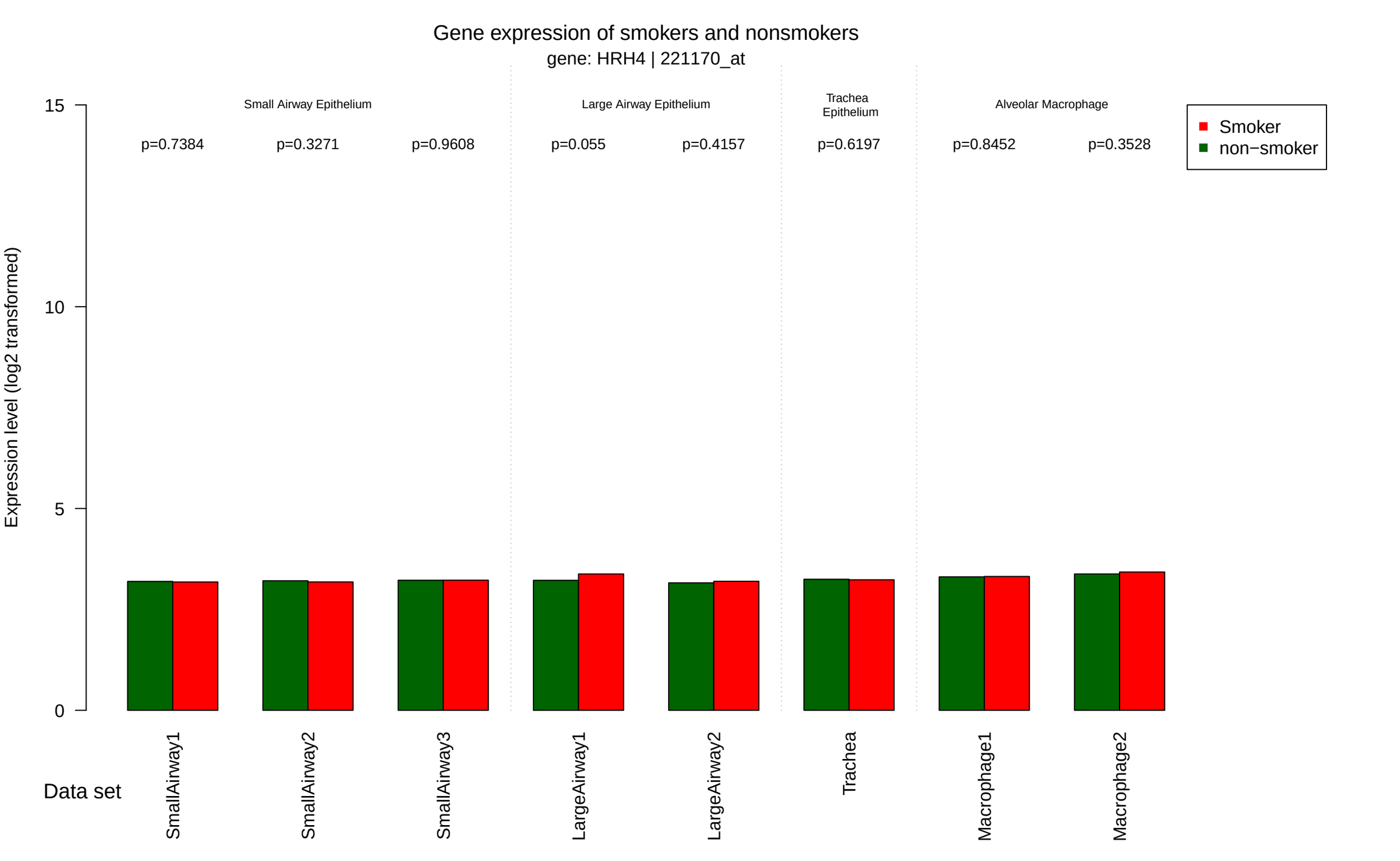
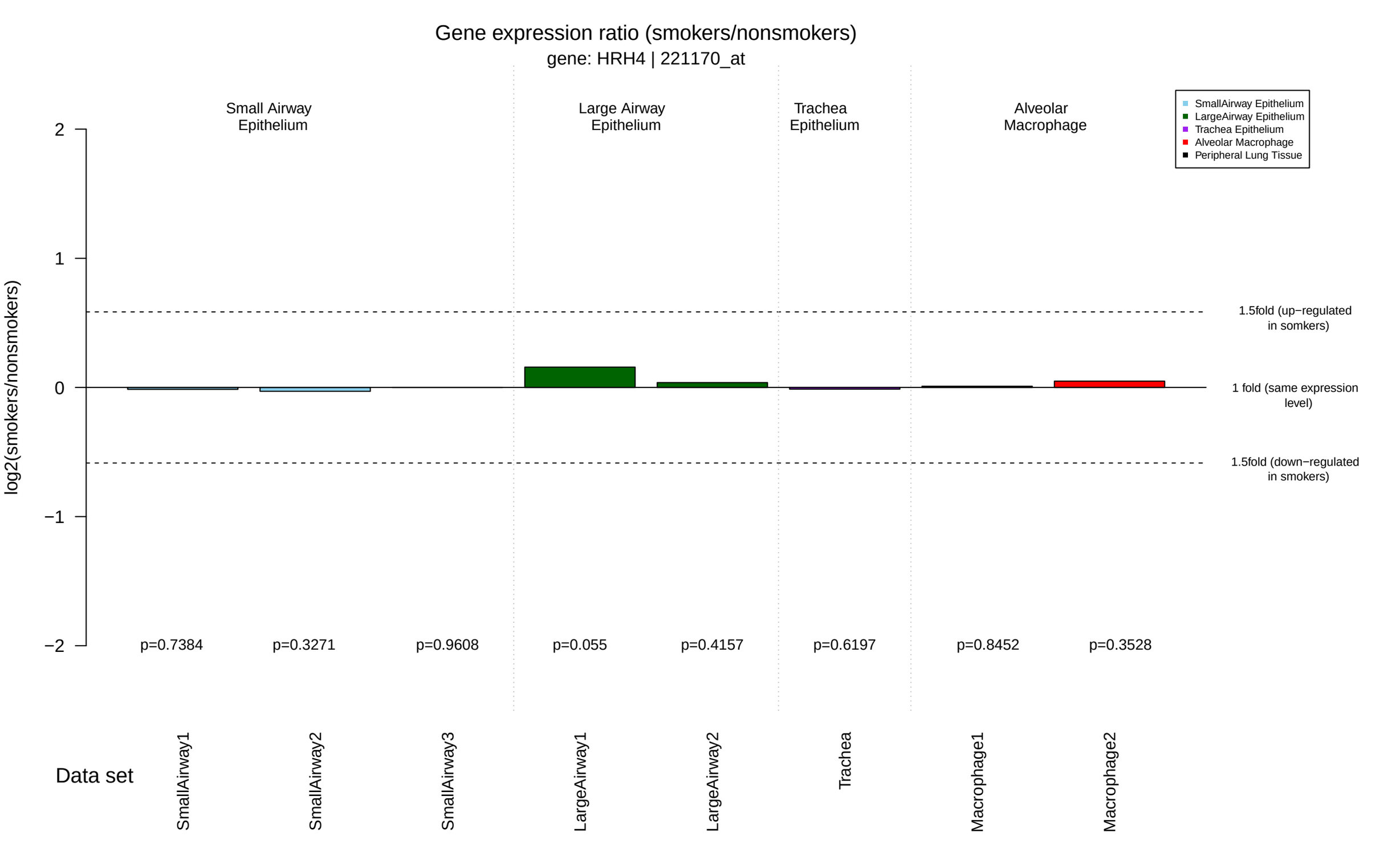
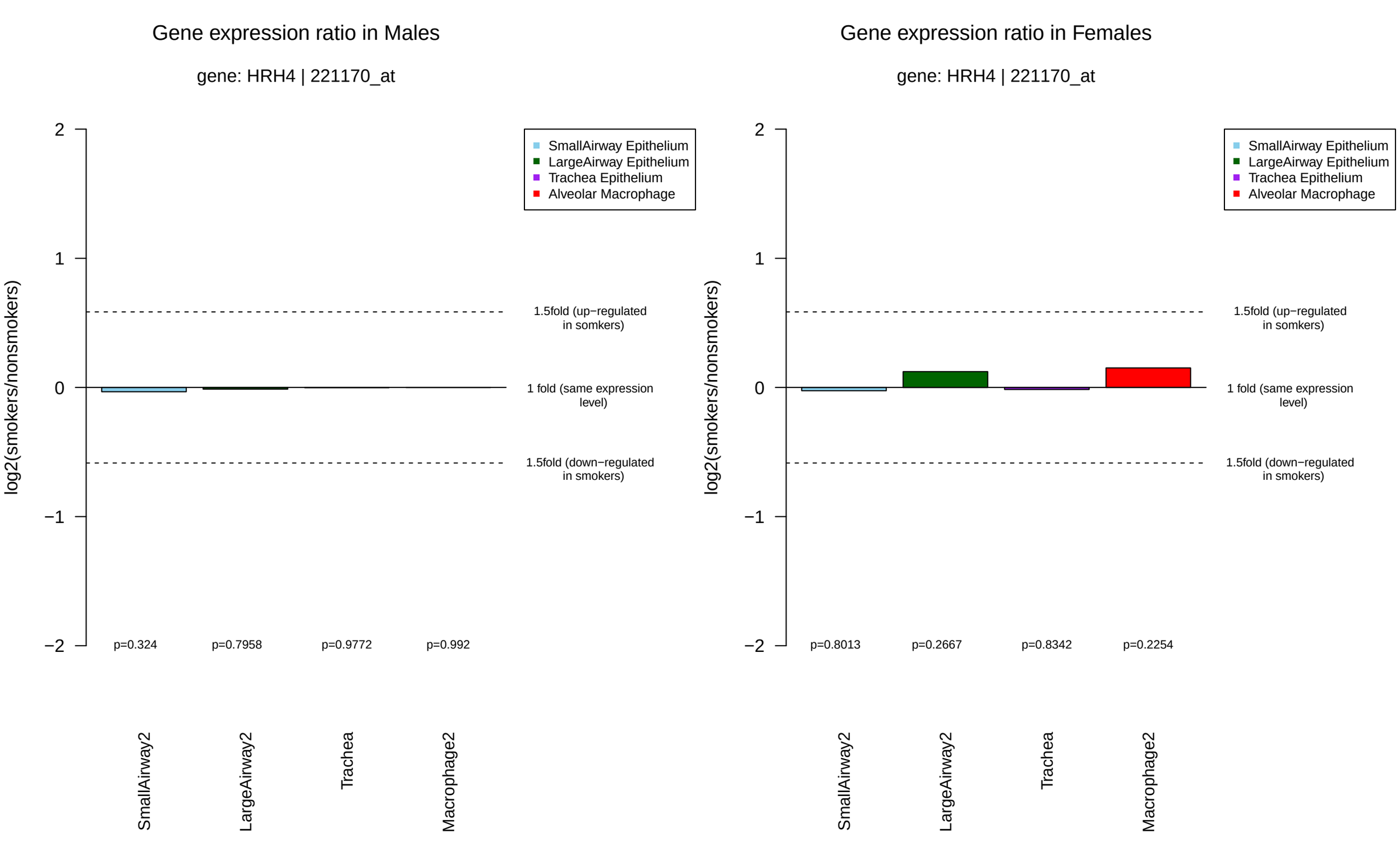
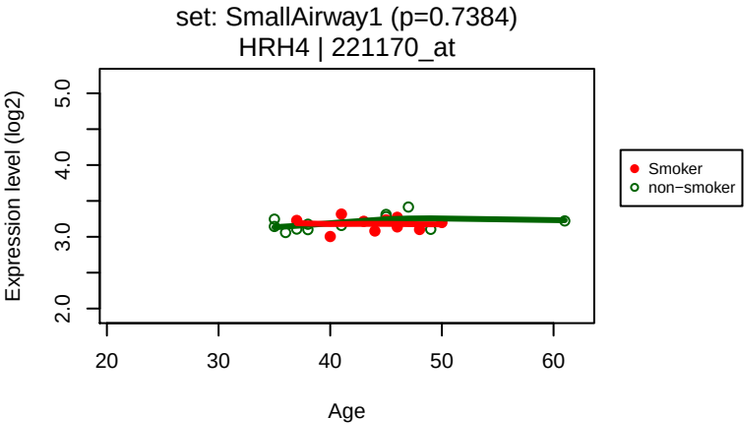
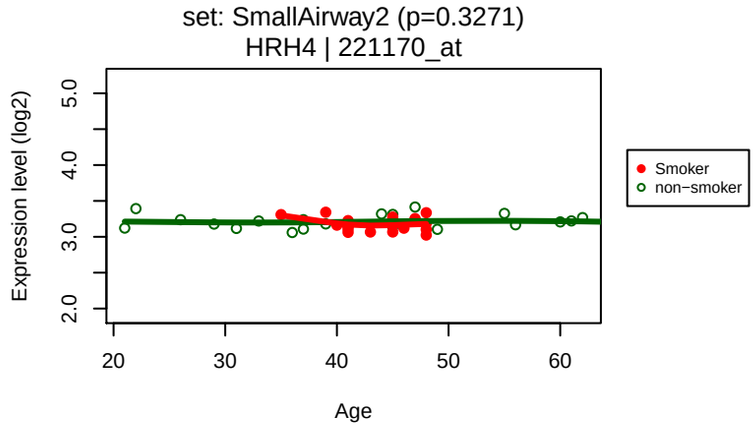
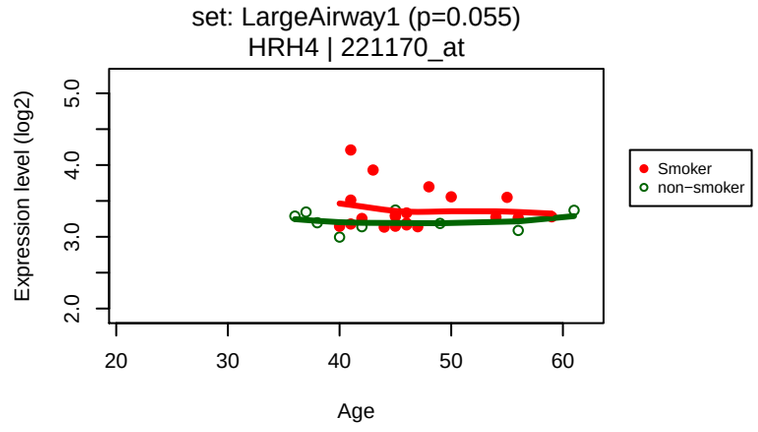

Gene expression of smokers and nonsmokers
gene: HRH4 | 221170_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7384
p=0.3271
p=0.9608
p=0.055
p=0.4157
p=0.6197
p=0.8452
p=0.3528
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

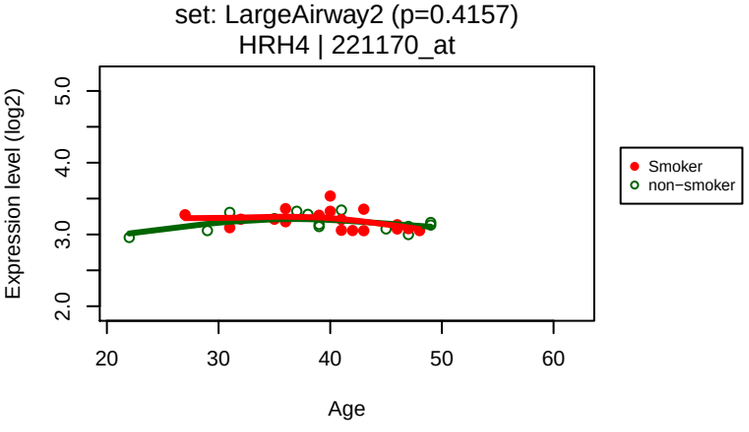

Gene expression ratio (smokers/nonsmokers)
gene: HRH4 | 221170_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7384
p=0.3271
p=0.9608
p=0.055
p=0.4157
p=0.6197
p=0.8452
p=0.3528
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

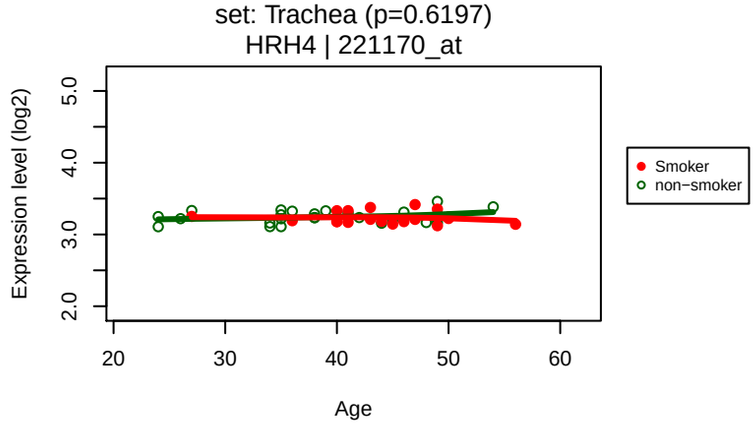

Gene expression ratio in Males
gene: HRH4 | 221170_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.324
p=0.7958
p=0.9772
p=0.992
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HRH4 | 221170_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8013
p=0.2667
p=0.8342
p=0.2254
SmallAirway2
LargeAirway2
Trachea
Macrophage2

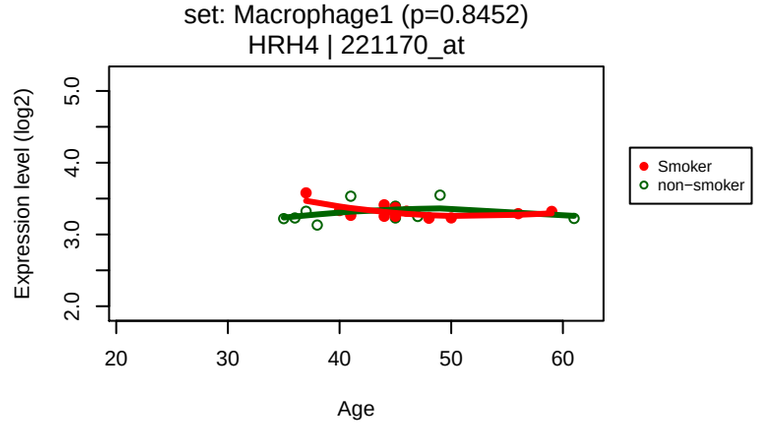

set: SmallAirway1 (p=0.7384)
HRH4 | 221170_at
Expression level (log2)
Age
Smoker
non−smoker

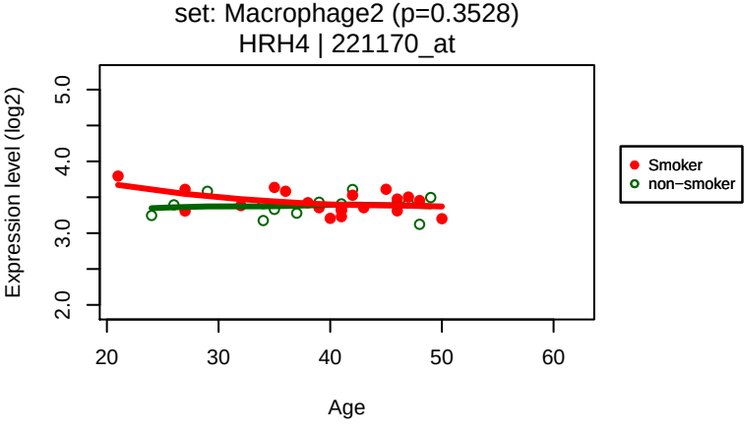

set: SmallAirway2 (p=0.3271)
HRH4 | 221170_at
Expression level (log2)
Age
Smoker
non−smoker

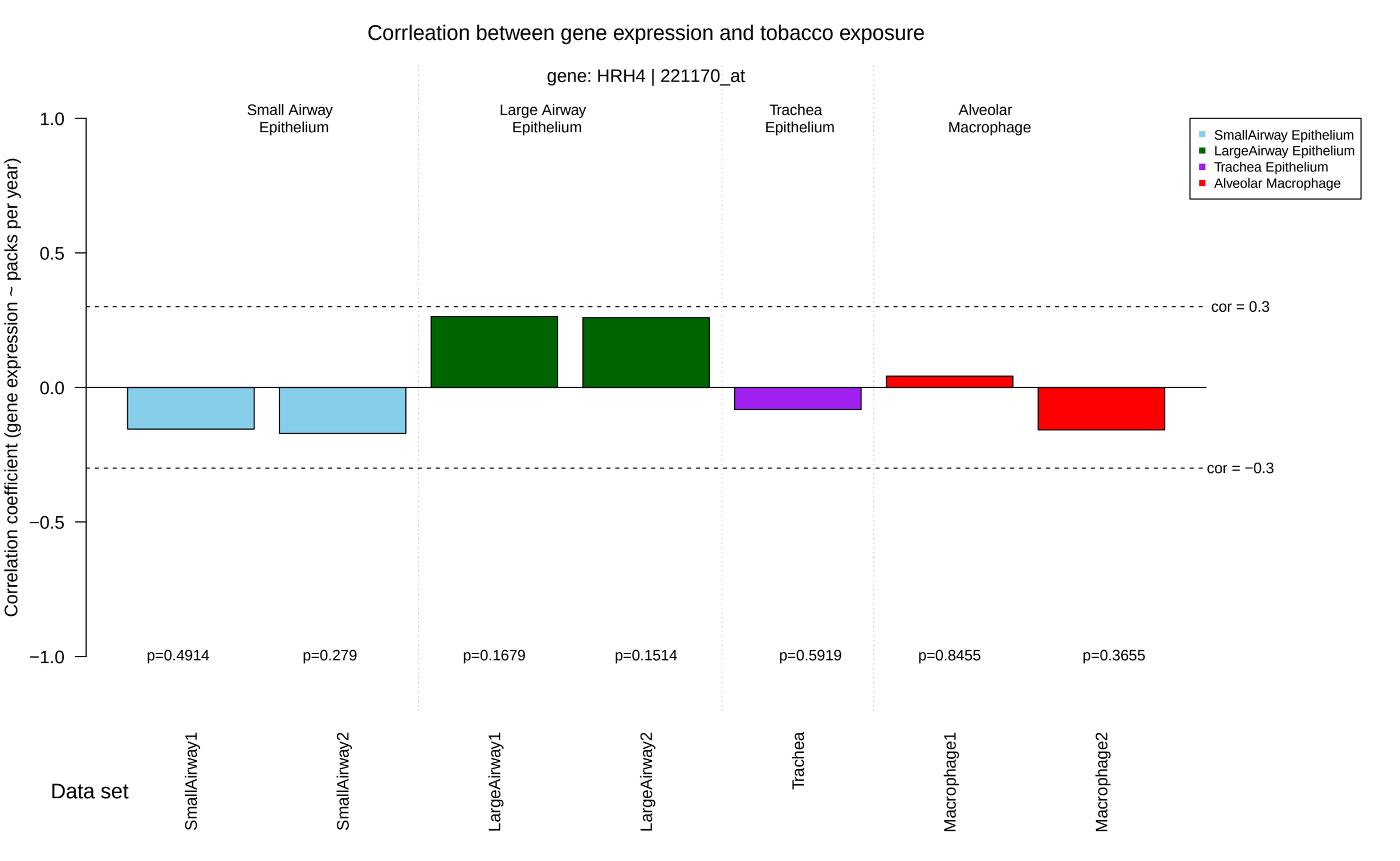

set: LargeAirway1 (p=0.055)
HRH4 | 221170_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4157)
HRH4 | 221170_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6197)
HRH4 | 221170_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8452)
HRH4 | 221170_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3528)
HRH4 | 221170_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HRH4 | 221170_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4914
p=0.279
p=0.1679
p=0.1514
p=0.5919
p=0.8455
p=0.3655
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2